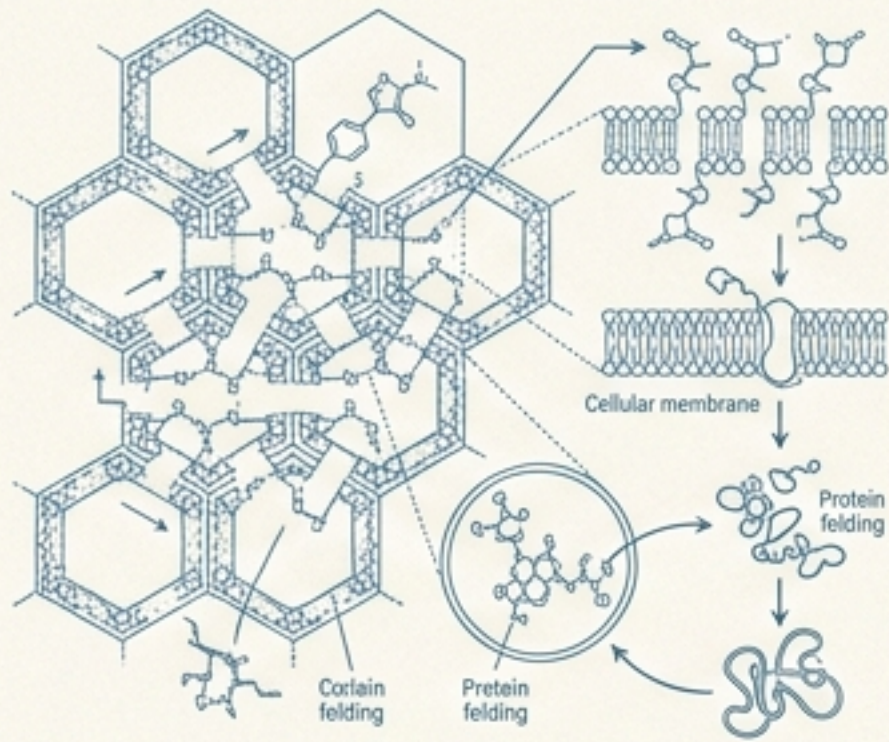


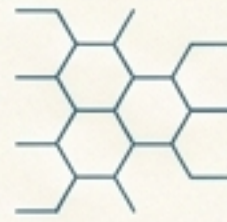
Morphology as Computation

Topology, Criticality, and
Structured Intelligence

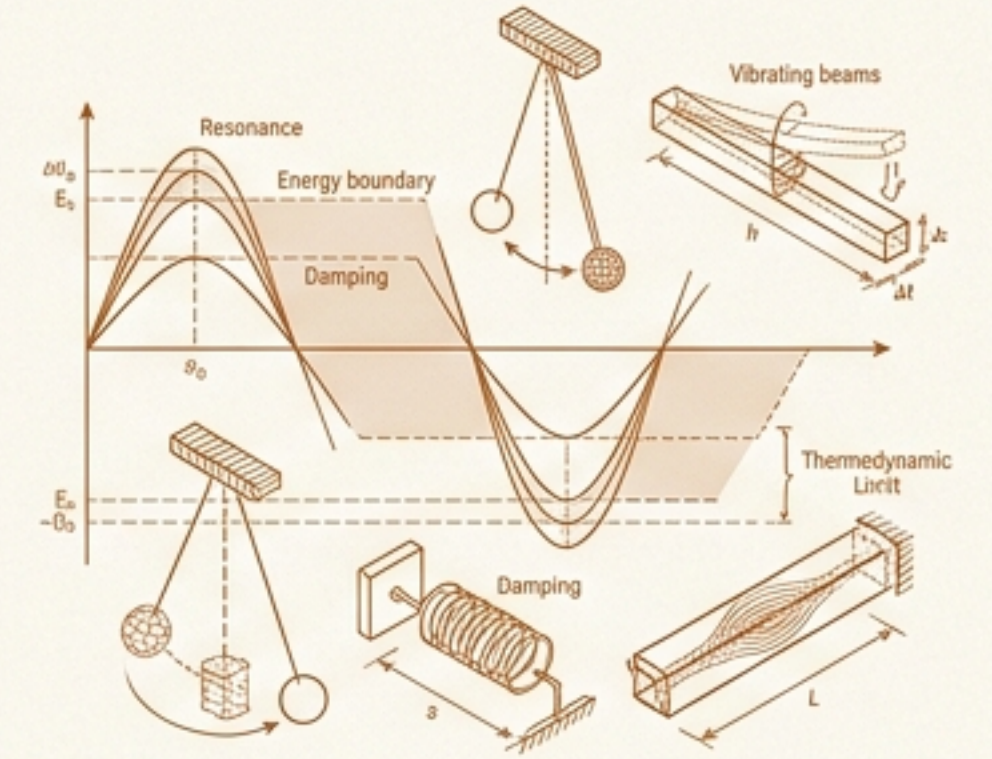
Flyxion | Working Paper, 2026



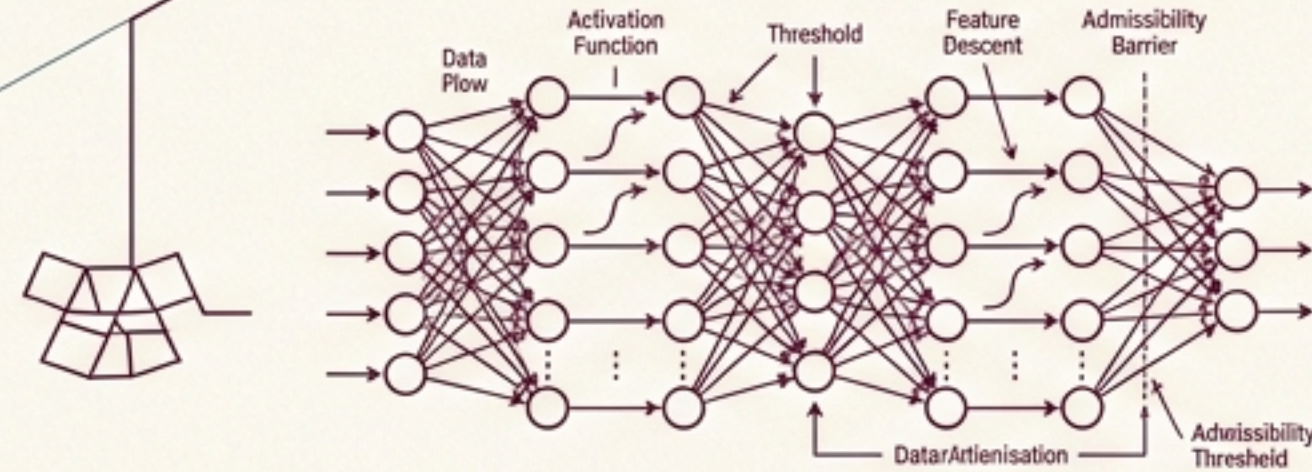
Biology (Self-assembly)



**Organization does not
require instructions.
It requires constraints.**



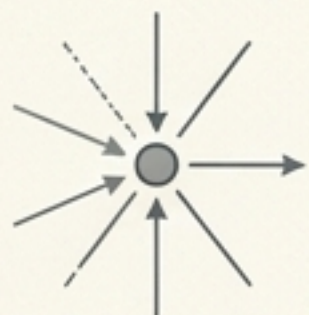
Physics (Optomechanics)



Artificial Systems (AI Inference)

What a system is at any moment is less fundamental than what transitions it can sustain without losing coherence.

Content-Primary (Localizationist)



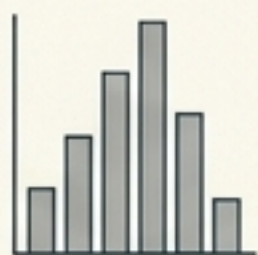
Causality: Localized.

A symptom or behavior lives at a specific, damaged node.



Structure: Material.

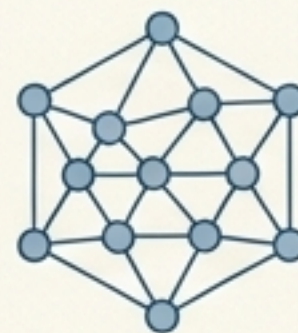
Morphology is an implementation detail; matter executes algorithms.



AI Strategy: Unconstrained Statistics.

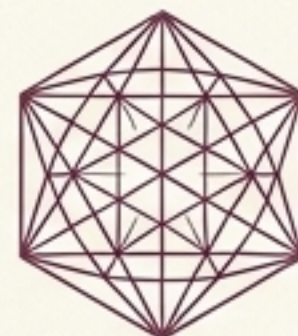
Pure aggregation over uniform feature distributions.

Constraint-Primary (Admissibility)



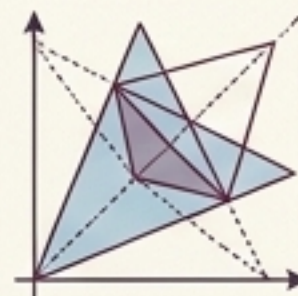
Causality: Distributed Manifold.

Organization is a topological property of the entire field.



Structure: Geometric.

Substrate morphology is the primary computational mechanism.

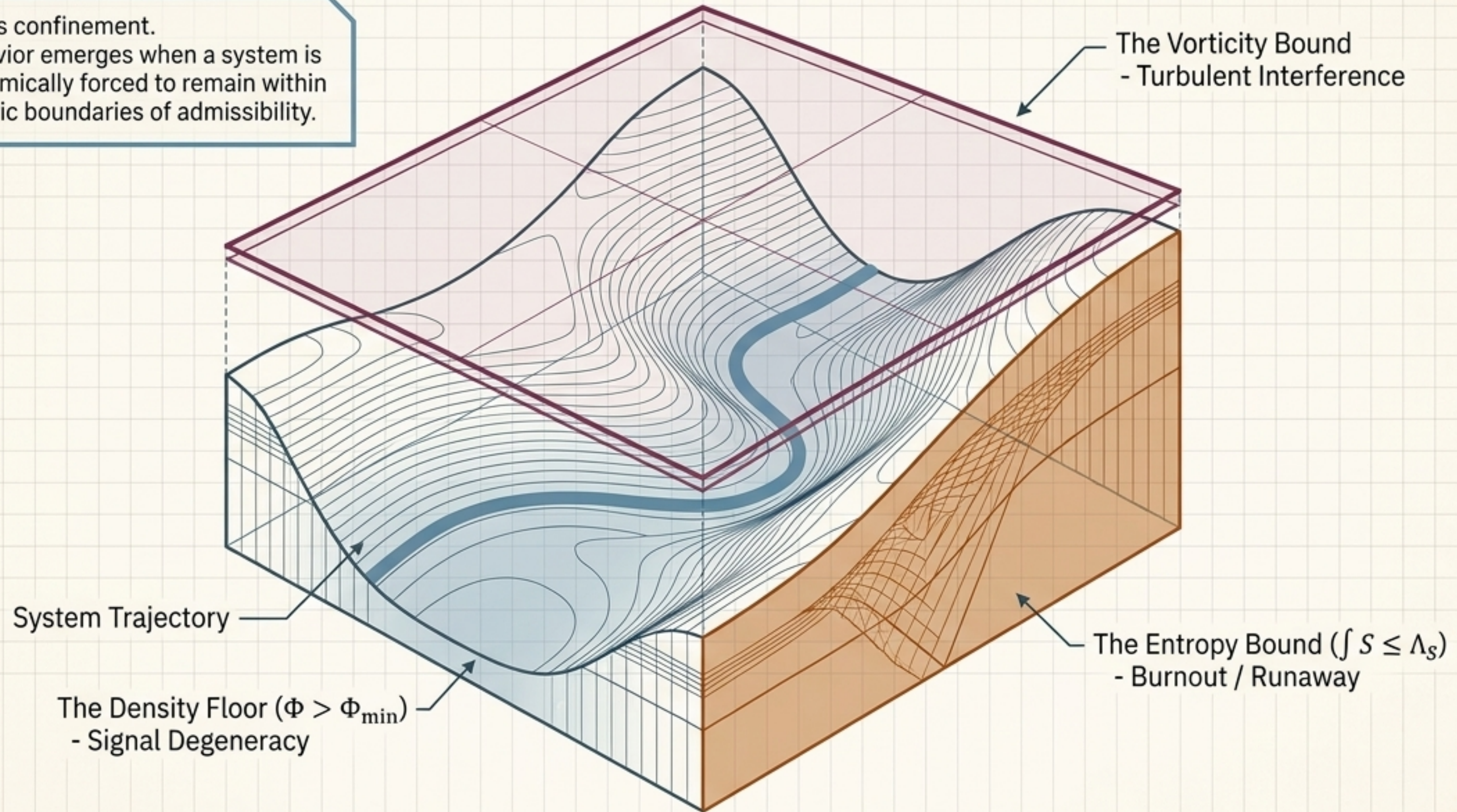


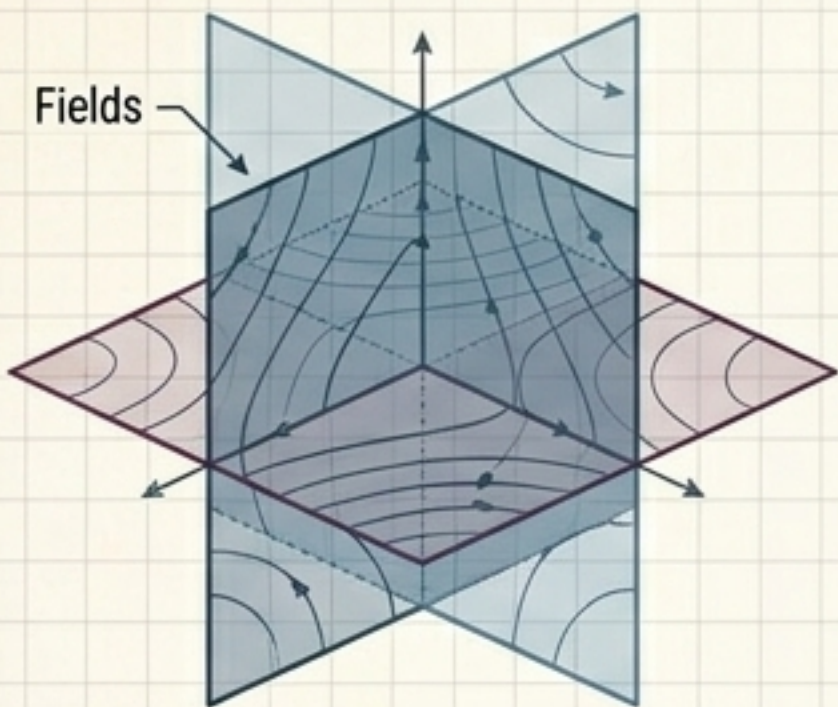
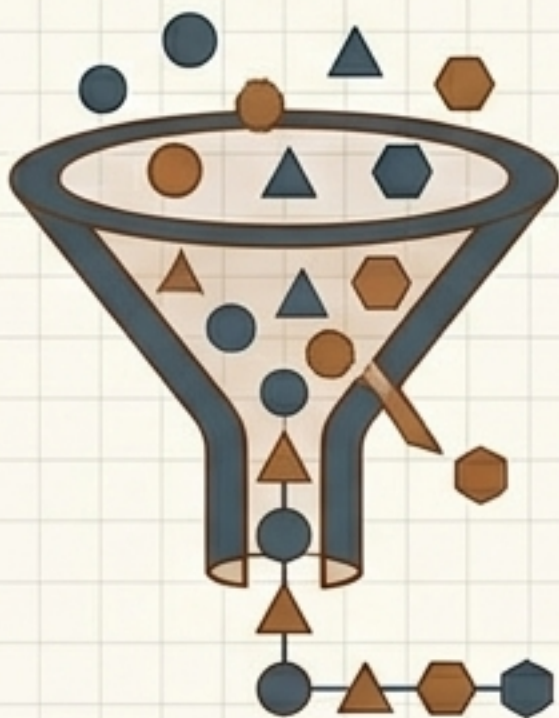
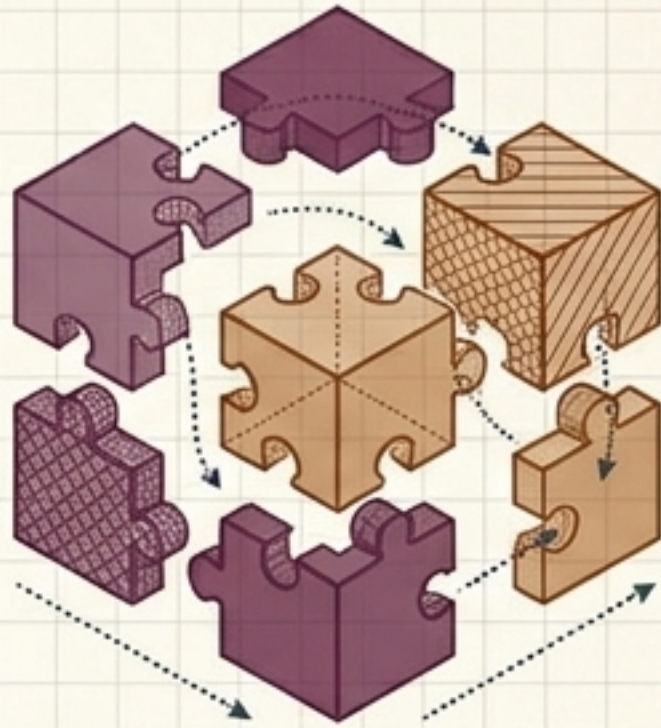
AI Strategy: Structural Priors.

Hypothesis spaces are geometrically filtered before statistical processing.

THE ADMISSIBILITY MANIFOLD

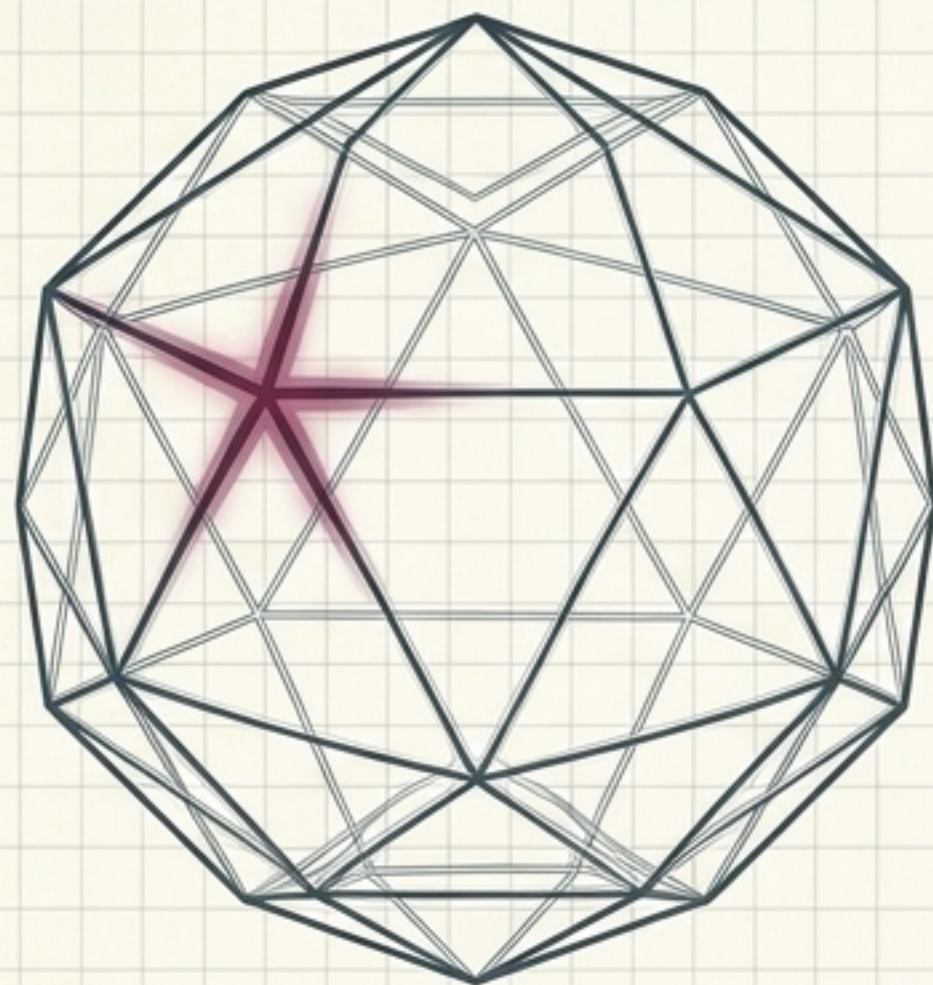
Coherence is confinement.
Global behavior emerges when a system is
thermodynamically forced to remain within
the geometric boundaries of admissibility.



1 A RSVP (Relativistic Scalar-Vector Plenum)	2 B CLIO (Constraint-Leveraged Inference)	3 C TARTAN (Trajectory-Aware Recursive Tiling)
 Domain: Physics & Biology Core Mechanism: Bounded field dynamics (Density Φ, Flow $\mathbf{v}$, Entropy S) Output: Macroscopic Coherence	 Domain: Epistemology & AI Core Mechanism: Structural priors over relational configurations Output: Intelligible Inference	 Domain: Computation & Memory Core Mechanism: Trajectory constraint-closure across fragmented records Output: Distributed Memory

RSVP in Action (Physics & Biology)

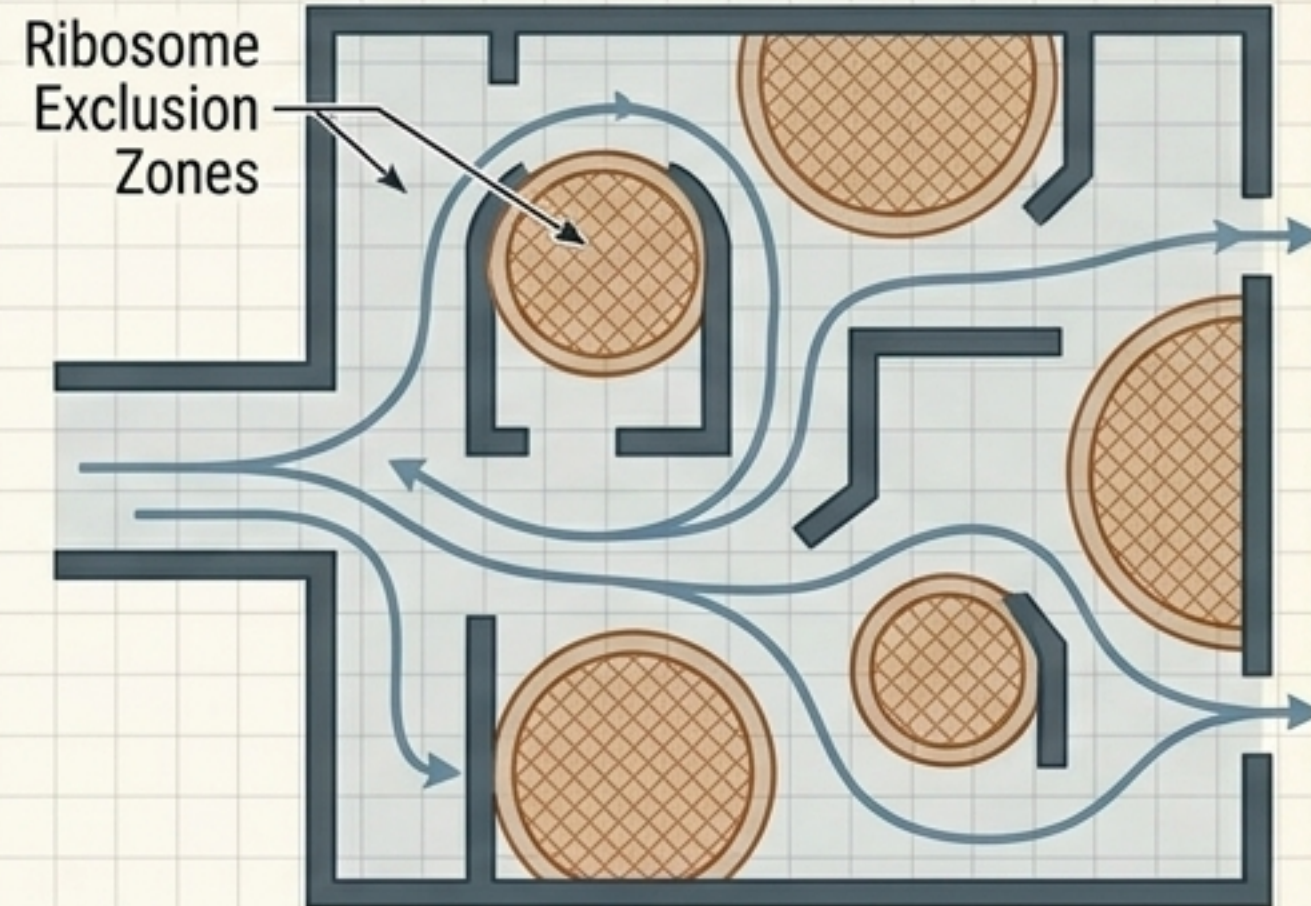
Protein Nanocages



Symmetry as Constraint Residue

Monomers have no explicit assembly instructions. Global polyhedral geometry is simply the thermodynamic residue of a global closure constraint minimizing free energy.

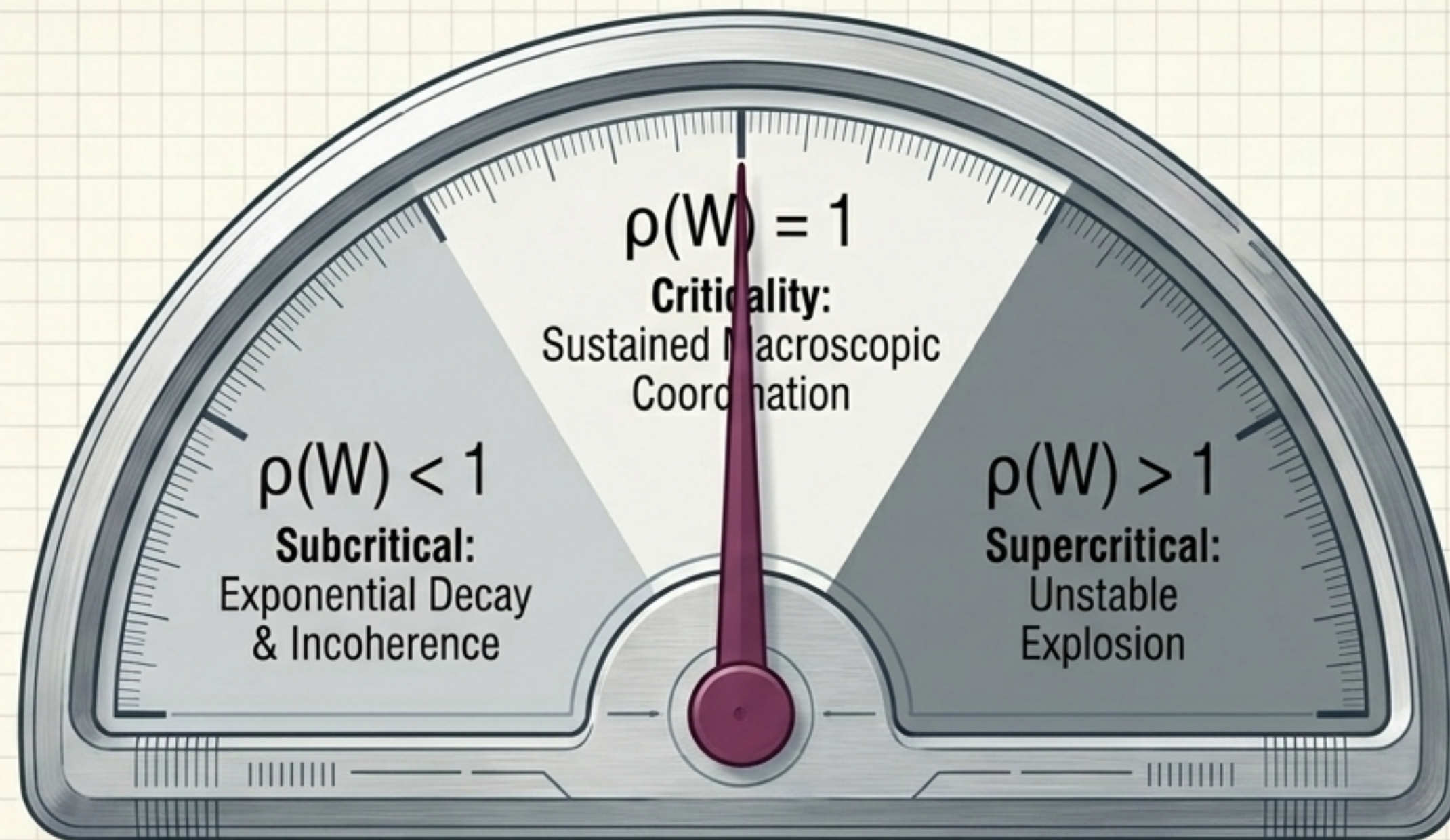
Intracellular Transport



Topology as Routing Infrastructure

The cell does not assign explicit addresses to vesicles. It builds a semantic infrastructure of compatibility predicates, where destinations emerge from intersection geometry.

Spectral Radius Dial

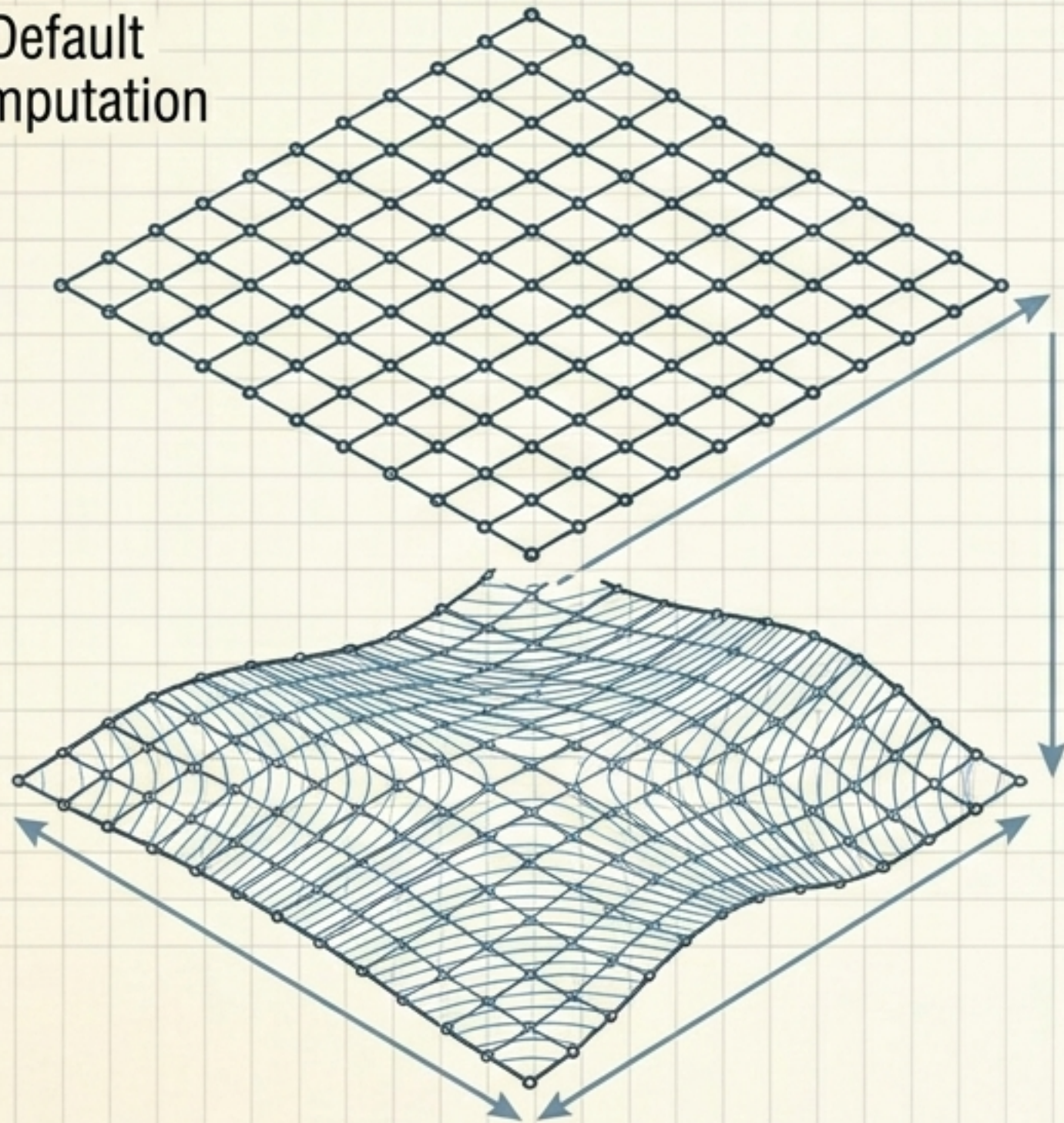


Randomly interacting biological neural networks only achieve large-scale power-law coherence when tuned precisely to $\rho(W) = 1$.

Criticality is not an empirical coincidence. It is the mathematical boundary condition of the admissibility manifold.

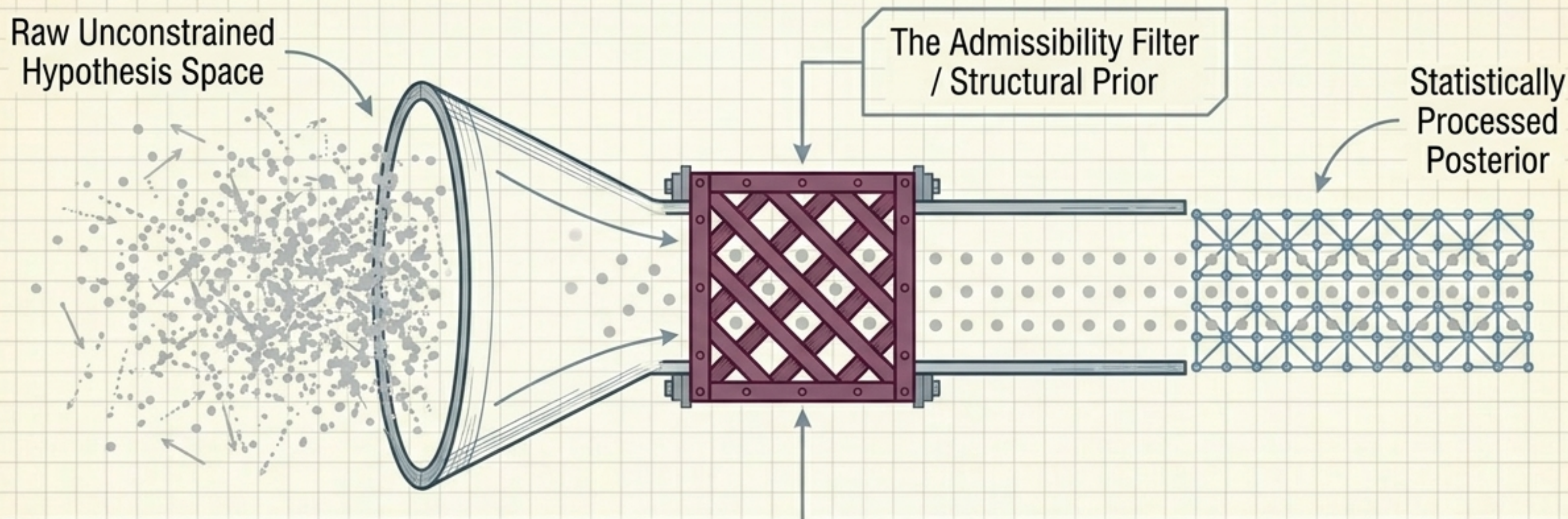
Xylomorphic Computation (Neuromorphic Matter)

Default
Computation



- Cognition is distributed across a deformable material substrate, not localized in a rigid processor.
- Substrate morphology is the computation. As the physical body bends, the system's admissibility manifold mathematically deforms, shifting its computational capacity.
- The effective Hamiltonian is directly coupled to the substrate's spatial displacement field.

The CLIO Mechanism: Admissibility Filtering & Structural Priors



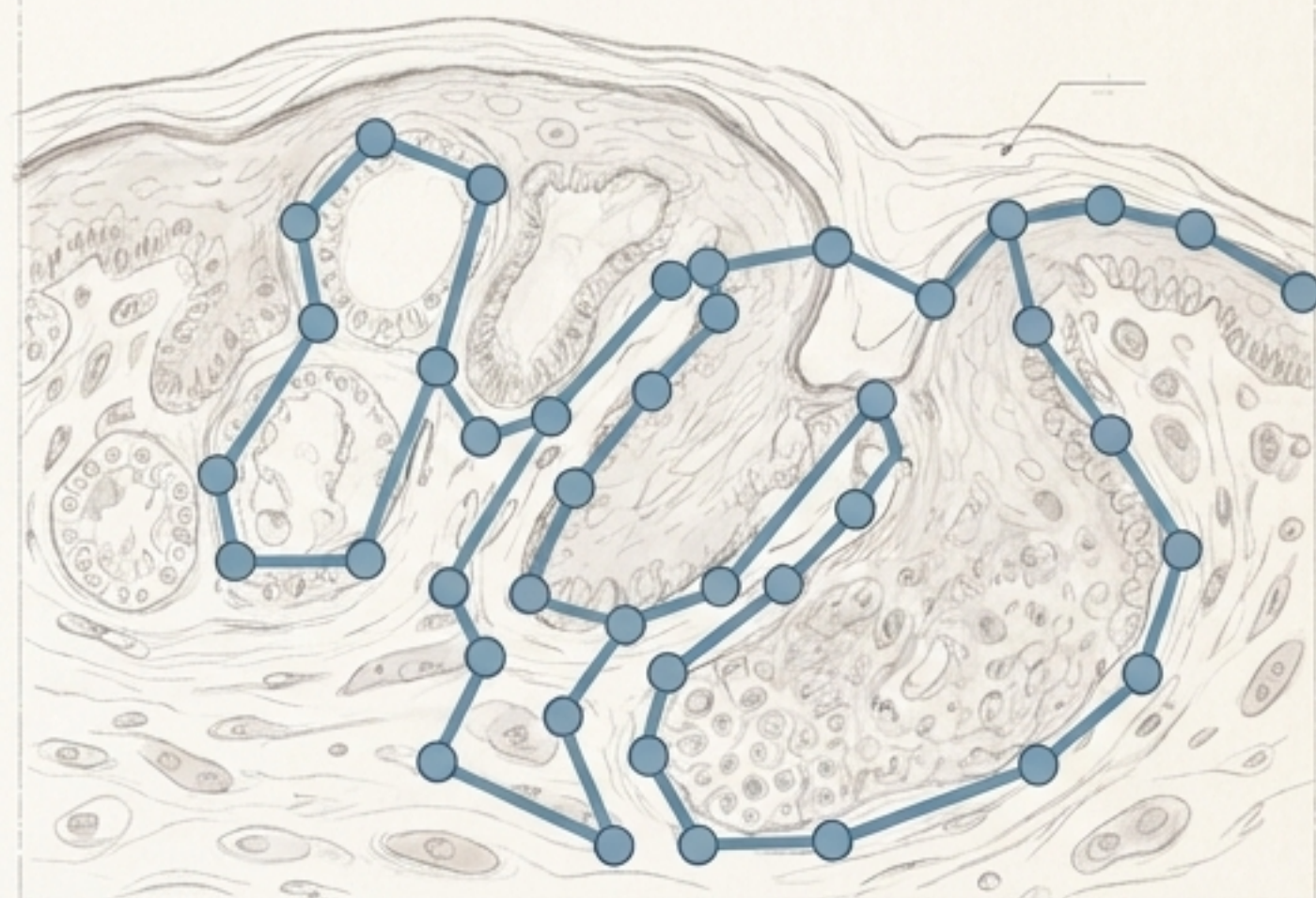
- The CLIO Mechanism: The prior acts as a hard filter, eliminating inferential noise from inadmissible regions before statistical likelihood is calculated.
- Evidence from Amharic News AI: Incorporating semantic ontology relationships (concept edges/axioms) heavily outperforms pure TF-IDF statistics.

Pure Statistical Aggregation



Aggregates local cellular features without spatial logic. Fails to capture the relational geometry pathologists use.

Topology-Aware GNN



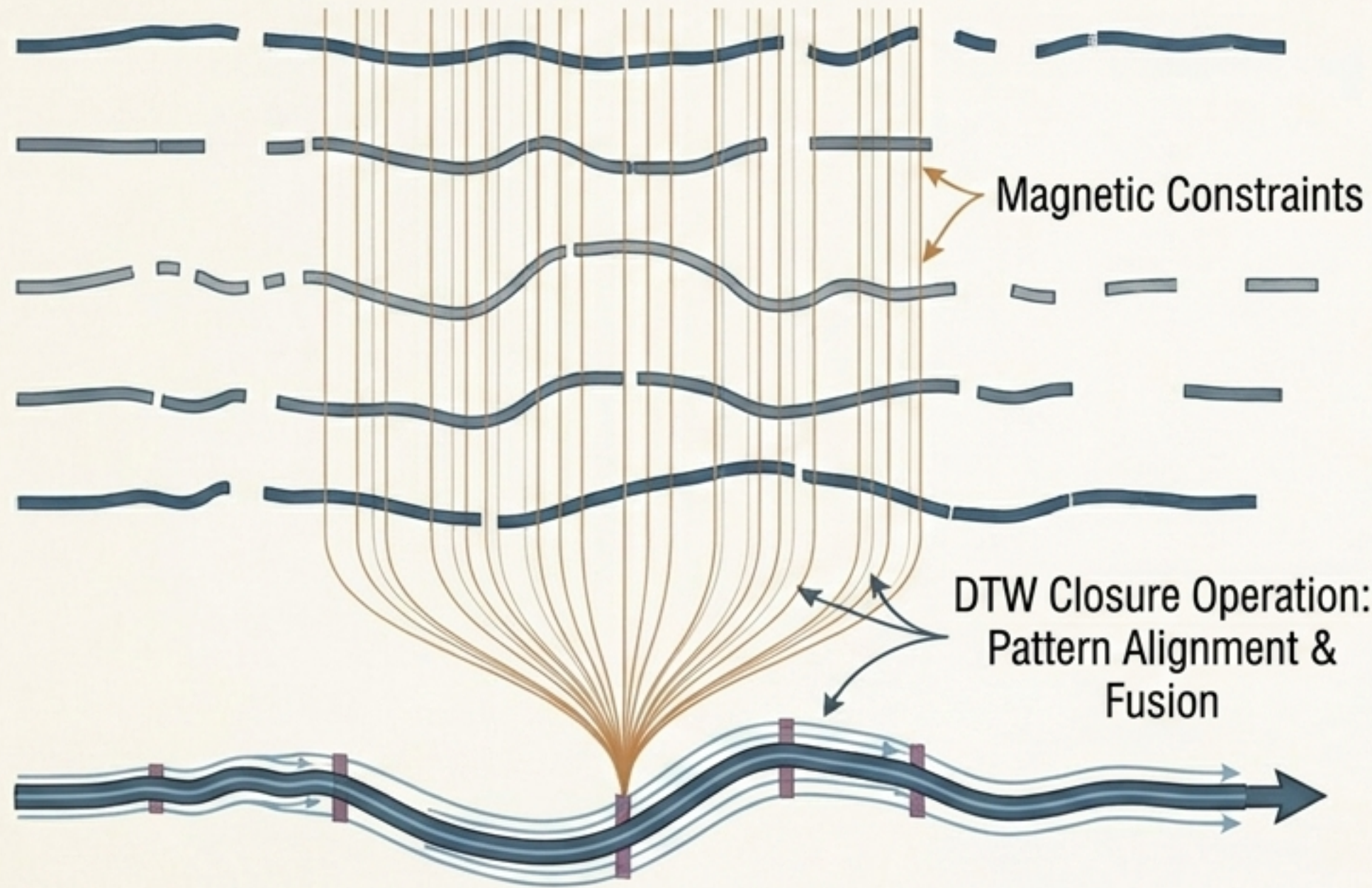
Message-passing is constrained strictly to P-admissible diagnostic substructures.

Clinical BCC AI

Dual Explanation architectures treat explanation as a structural trajectory constraint, not a post-hoc hallucination. Inference is geometrically locked to clinical ontologies.

Trajectory Tiling and Distributed Memory

Dynamic Time Warping



Continuous Paleoclimate Trajectory

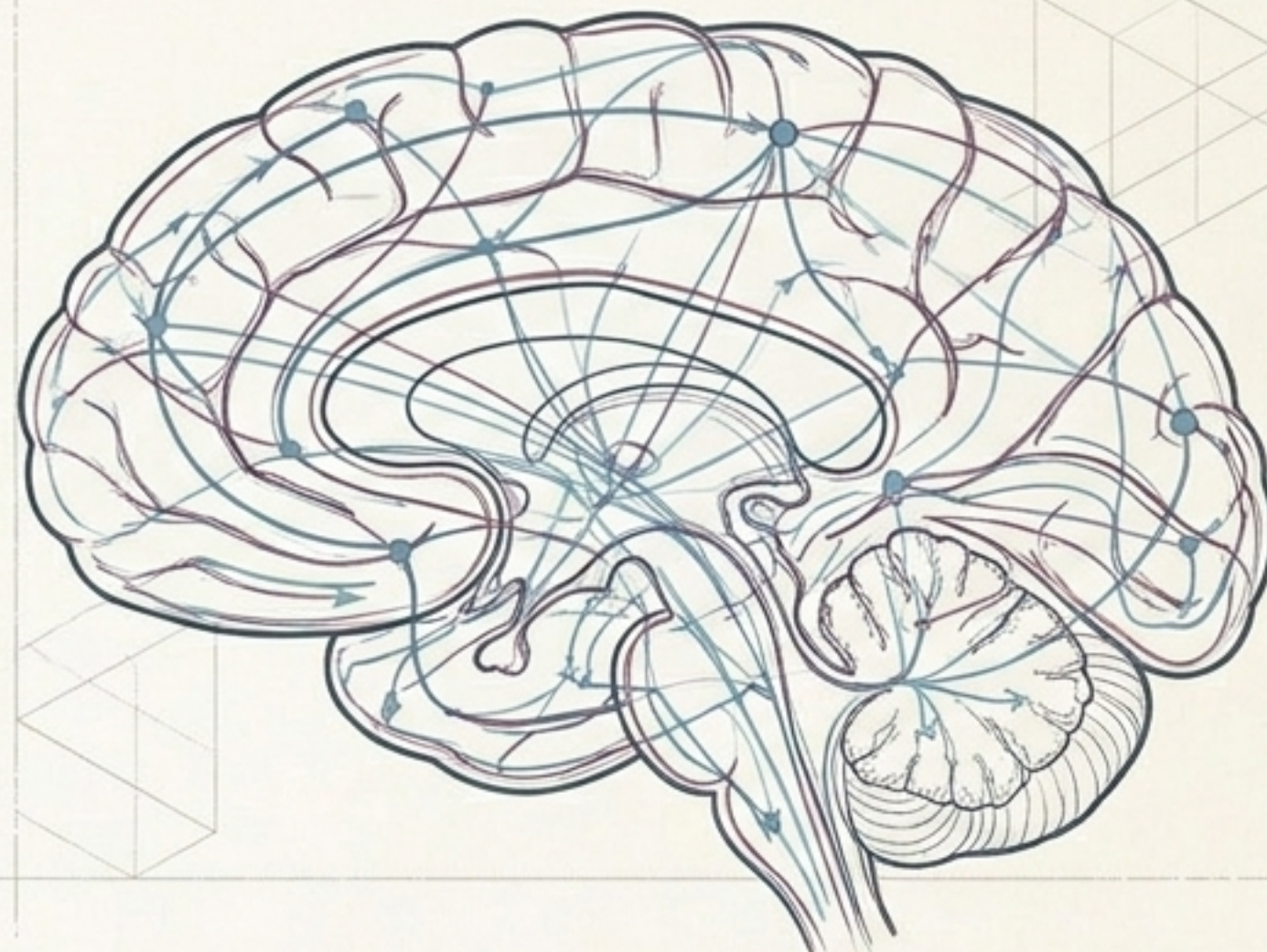
- Memory does not exist as discrete stored states in a single node. It is distributed structural residue.
- Dynamic Time Warping (DTW) acts as the TARTAN closure operation: it identifies the unique continuous trajectory that satisfies all partial tilings simultaneously.
- Result: Fragmented Red Sea rhodolith growth bands generate precise daily-resolution paleotemperature archives.

The Localizationist View



Symptom = Faulty Node.

RSVP View



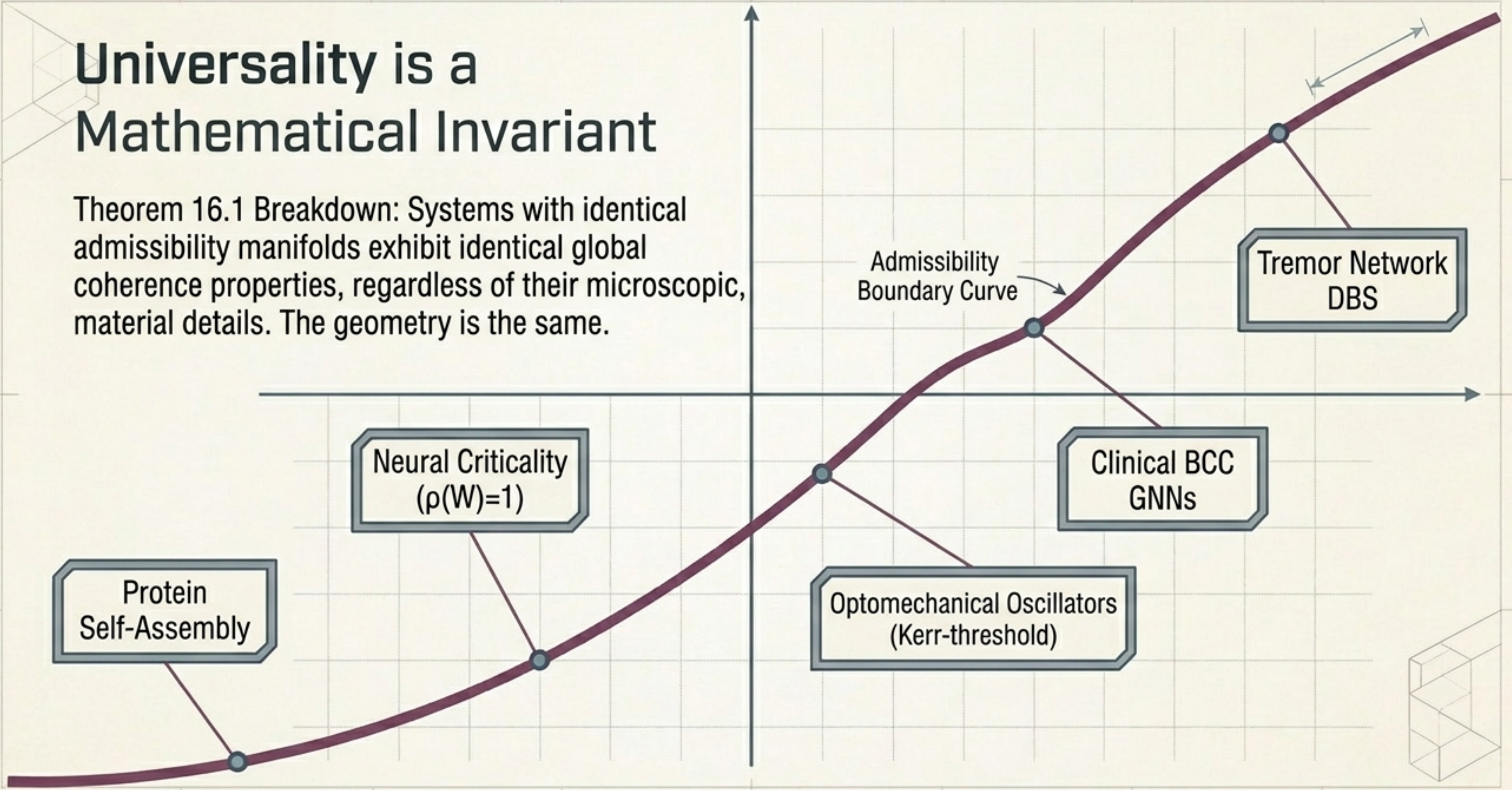
Symptom = State of the Network Field.

**Mechanism of Action:
Topological Modification**

Deep Brain Stimulation (DBS) for tremor does not work by suppressing a localized pathological signal. DBS is a topological modification. It displaces the entire network geometry outward, forcing the brain's functional state to exit the distributed causal manifold of the symptom.

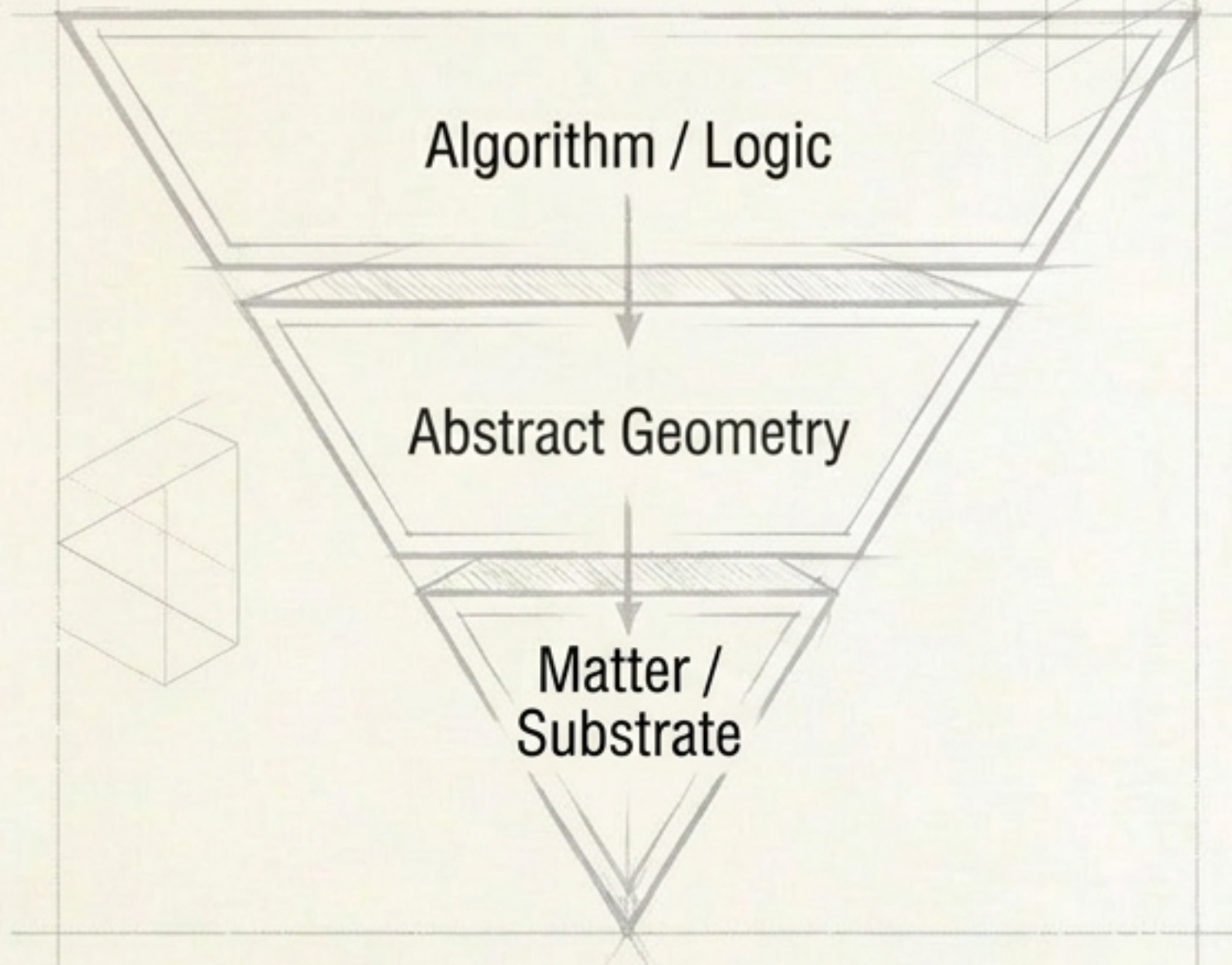
Universality is a Mathematical Invariant

Theorem 16.1 Breakdown: Systems with identical admissibility manifolds exhibit identical global coherence properties, regardless of their microscopic, material details. The geometry is the same.



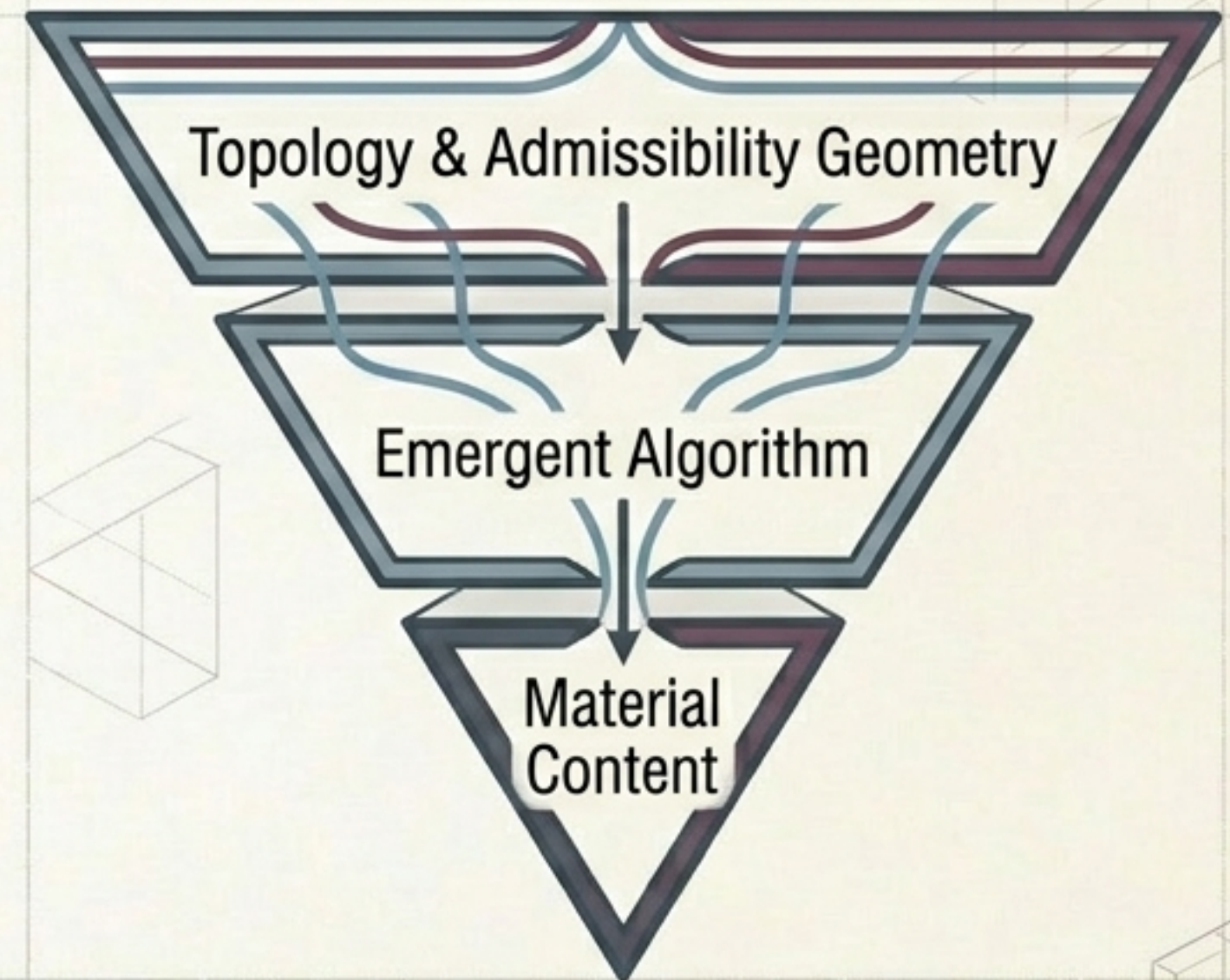
Implications: Reversing Computational Reductionism

The Old Hierarchy

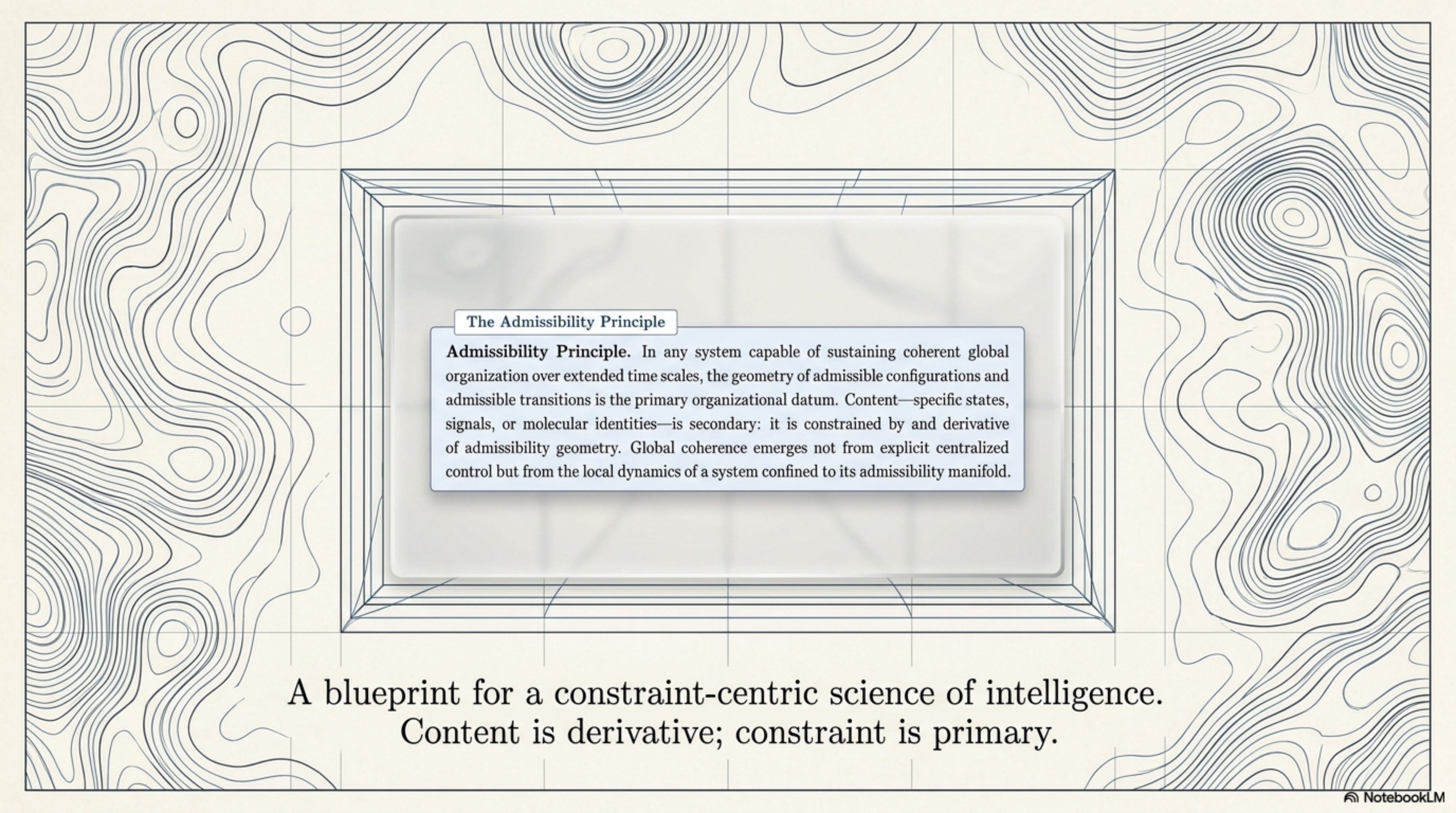


Computational Reductionism.
Morphology is just an implementation detail.

The New Hierarchy



Constraint-Centric Paradigm. Morphology
dictates what computations are even possible.



The Admissibility Principle

Admissibility Principle. In any system capable of sustaining coherent global organization over extended time scales, the geometry of admissible configurations and admissible transitions is the primary organizational datum. Content—specific states, signals, or molecular identities—is secondary: it is constrained by and derivative of admissibility geometry. Global coherence emerges not from explicit centralized control but from the local dynamics of a system confined to its admissibility manifold.

A blueprint for a constraint-centric science of intelligence.
Content is derivative; constraint is primary.